Activity above one half in finite Glauber annealing

Xiaojian Zeng*
PBC School of Finance, Tsinghua University, 100190
(Dated: September 12, 2026)

The equilibrium jump probability of a heat-bath Glauber update on a regular state graph is at most one half. We give a finite counterexample to extending this bound to arbitrary monotone cooling from a Gibbs initial state. An eleven-state, four-regular graph and a two-update cooling schedule at strictly positive, finite temperatures give a second-step activity greater than 63/125 and a two-step mean greater than 251/500. All sign decisions reduce to exact rational arithmetic. Both violations persist when each state energy is independently changed by at most $\epsilon/1000$, with ϵ the specified energy unit and the initial Gibbs state redefined accordingly. The mechanism is transient overoccupation of an intermediate energy level. Instantaneous detailed balance and monotone cooling therefore do not suffice for a universal one-half activity bound on regular state graphs.

I. MODEL AND THE ACTIVITY QUESTION

The activity of a discrete thermal update is the probability of changing microstate per proposal. Luo *et al.* considered this observable in bounds on annealing performance [1, 2]. Their detailed treatment establishes an equilibrium identity and conjectures the bound $\mathcal{A}_k \le 1/2$ along cooling trajectories. Here the precise reference is Eq. (39) of arXiv:2311.10424v1; its dynamics and activity are Eqs. (30)–(33). Equation and page references to the two Luo papers below refer to their v1 author preprints. The counterexample concerns the general regular-state-graph formulation, with finite temperature changes allowed.

Let $G = (V, F)$ be a finite, connected, undirected simple graph of degree z, and assign a fixed finite energy E_i to each vertex. Set $k_B = 1$. With the destination index first, the transition matrix is

$$P_{ij}(\beta) = \begin{cases} \dfrac{1}{z[1+e^{\beta(E_i-E_j)}]}, & \{i,j\} \in F, \\ 1 - \displaystyle\sum_{l\neq j} P_{lj}(\beta), & i = j, \\ 0, & \text{otherwise.} \end{cases} \tag{1}$$

One neighbor is proposed uniformly, followed by the heat-bath acceptance decision. The diagonal collects rejections. Every matrix is column stochastic and obeys detailed balance with

$$\gamma_i(\beta) = \frac{e^{-\beta E_i}}{Z(\beta)}, \qquad Z(\beta) = \sum_{j\in V} e^{-\beta E_j}. \tag{2}$$

Indeed, on an edge the equilibrium directed jump mass is $e^{-\beta(E_i+E_j)}/\{zZ(e^{-\beta E_i}+e^{-\beta E_j})\}$, which is symmetric in i, j.

For $0 < \beta_0 \le \cdots \le \beta_K < \infty$, define

$$p^{(0)} = \gamma(\beta_0), \qquad p^{(k+1)} = P(\beta_{k+1})p^{(k)}, \tag{3}$$

$$\mathcal{A}_k = \sum_j p_j^{(k)} \sum_{i\neq j} P_{ij}(\beta_{k+1}), \qquad \overline{\mathcal{A}}_K = \frac{1}{K}\sum_{k=0}^{K-1} \mathcal{A}_k. \tag{4}$$

There is no reequilibration between these single-proposal updates. Intermediate distributions are neither assumed Gibbsian nor assumed ordered by energy. Activity counts all microstate changes, including equal-energy transitions.

To see the distinction from equilibrium, write $w_i = e^{-\beta E_i}$. Combining the two directions of each undirected edge gives

$$\begin{aligned} \mathcal{A}(p,\beta) &= \frac{1}{z}\sum_{\{i,j\}\in F} \frac{w_i p_j + w_j p_i}{w_i + w_j} \\ &= \frac{1}{2} - \frac{1}{2z}\sum_{\{i,j\}\in F} \frac{(w_i - w_j)(p_i - p_j)}{w_i + w_j}. \end{aligned} \tag{5}$$

The second line uses $\sum_{\{i,j\}\in F}(p_i + p_j) = z$. At $p = \gamma(\beta)$ it becomes

$$\mathcal{A}_\gamma = \frac{1}{2} - \frac{1}{2z}\sum_{\{i,j\}\in F} \frac{(\gamma_i - \gamma_j)^2}{\gamma_i + \gamma_j} \le \frac{1}{2}. \tag{6}$$

These are the undirected-edge versions of Eqs. (40) and (38) of Ref. [1], whose neighborhood sums count each edge twice. At nonequilibrium the correction has no definite sign. For example, ordering p and γ alike on every edge suffices to make it nonnegative, but this ordering need not be preserved by cooling.

* xiaojianzeng@foxmail.com

II. AN ELEVEN-STATE FINITE COUNTEREXAMPLE

Divide the vertices into

$$\begin{aligned} A &= \{a_0, a_1, a_2\}, \\ B &= \{b_0, b_1, b_2, b_3\}, \\ C &= \{c_0, c_1, c_2, c_3\}. \end{aligned} \tag{7}$$

Join every A vertex to every B vertex, join b_j to c_j, and join every pair of distinct C vertices. There are no other edges. The degrees in the three classes are 4, $3+1$, and $1+3$, respectively. Thus the graph is simple, connected, and four-regular, with 22 undirected edges.

Fix $\epsilon > 0$ and take

$$E_{a_i} = 0, \qquad E_{b_j} = \epsilon, \qquad E_{c_j} = 64\epsilon. \tag{8}$$

With $b_k = \beta_k \epsilon$ the dimensionless inverse temperatures, use

$$(e^{b_0}, e^{b_1}, e^{b_2}) = \left(\frac{10001}{10000}, \frac{33}{32}, 64\right). \tag{9}$$

All three temperatures are finite and strictly positive, and $0 < b_0 < b_1 < b_2$. There are exactly two updates, so $K = 2$. The result is

$$\boxed{\mathcal{A}_1 > \frac{63}{125} > \frac{1}{2}, \qquad \overline{\mathcal{A}}_2 > \frac{251}{500} > \frac{1}{2}.} \tag{10}$$

A. Exact propagation and microscopic jump counting

Set

$$\begin{aligned} r &= \frac{10000}{10001}, \qquad s = r^{64}, \\ Z &= 3 + 4r + 4s, \qquad (u, v, w) = \frac{(1, r, s)}{Z}. \end{aligned} \tag{11}$$

The variables u, v, w are initial probabilities *per vertex*; the class probabilities are $3u, 4v, 4w$. Every vertex in a class has the same energy and the same number of neighbors in each class. Consequently the per-vertex probabilities remain equal within each class under an update.

At the first update, the acceptance probabilities for $B \to A$ and $C \to B$ are

$$h = \frac{33}{65}, \qquad \ell = \frac{33^{63}}{33^{63} + 32^{63}}. \tag{12}$$

The reverse acceptances are their complements. The net probability currents per edge toward lower energy are

$$J = \frac{hv - (1-h)u}{4}, \qquad L = \frac{\ell w - (1-\ell)v}{4}. \tag{13}$$

All currents use the same pre-update distribution. The resulting per-vertex probabilities are therefore

$$x = u + 4J = h(u+v), \qquad y = v - 3J + L, \qquad t = w - L. \tag{14}$$

They obey $3x + 4y + 4t = 1$ and are positive. In particular,

$$\begin{aligned} y &= \frac{3(1-h)u + (3 + \ell - 3h)v + \ell w}{4}, \\ t &= \frac{(1-\ell)v + (4-\ell)w}{4}. \end{aligned} \tag{15}$$

The first-step activity is

$$\mathcal{A}_0 = 3u(1-h) + v(3h + 1 - \ell) + w(3/2 + \ell). \tag{16}$$

The $3w/2$ term comes from the six edges inside C. Their net class current vanishes, but equal-energy proposals are accepted with probability $1/2$ and do change microstate. Discarding them when grouping states by energy would change the observable.

At the second update let $\rho = (1 + 64^{63})^{-1}$. The escape probabilities from an individual vertex are

$$a_A = \frac{1}{65}, \quad a_B = \frac{48}{65} + \frac{\rho}{4}, \quad a_C = \frac{5}{8} - \frac{\rho}{4}. \tag{17}$$

Thus

$$\mathcal{A}_1 = \frac{3x}{65} + \left(\frac{192}{65} + \rho\right) y + \left(\frac{5}{2} - \rho\right) t. \tag{18}$$

Equations (11)–(18) contain only integer powers and rational operations. Their exact evaluation gives Table I; Appendix A supplies a short executable verification. The lower endpoints for $\mathcal{A}_1$ and $\overline{\mathcal{A}}_2$ exceed $63/125$ and $251/500$, respectively, proving Eq. (10). No limiting temperature or floating-point tolerance enters this sign decision.

TABLE I. Exact rational enclosures. For every listed quantity X, the integer m gives $m/D \le X < (m+1)/D$, where $D = 10^{12}$. These are interval endpoints checked by integer arithmetic, not rounded equalities. The last three rows are probabilities per microstate after the first update.

Quantity X	Lower-endpoint numerator m
$\mathcal{A}_0$	499785676488
$\mathcal{A}_1$	504349849005
$\overline{\mathcal{A}}_2$	502067762747
x	92521064942
y	106989212896
t	73619988396

III. UNIFORM STABILITY UNDER ENERGY PERTURBATIONS

The violation does not require exact degeneracies. Keep the graph and temperatures fixed, replace E_i by $E_i + \epsilon d_i$, and impose

$$|d_i| \le \delta = \frac{1}{1000} \qquad \text{independently for every } i. \tag{19}$$

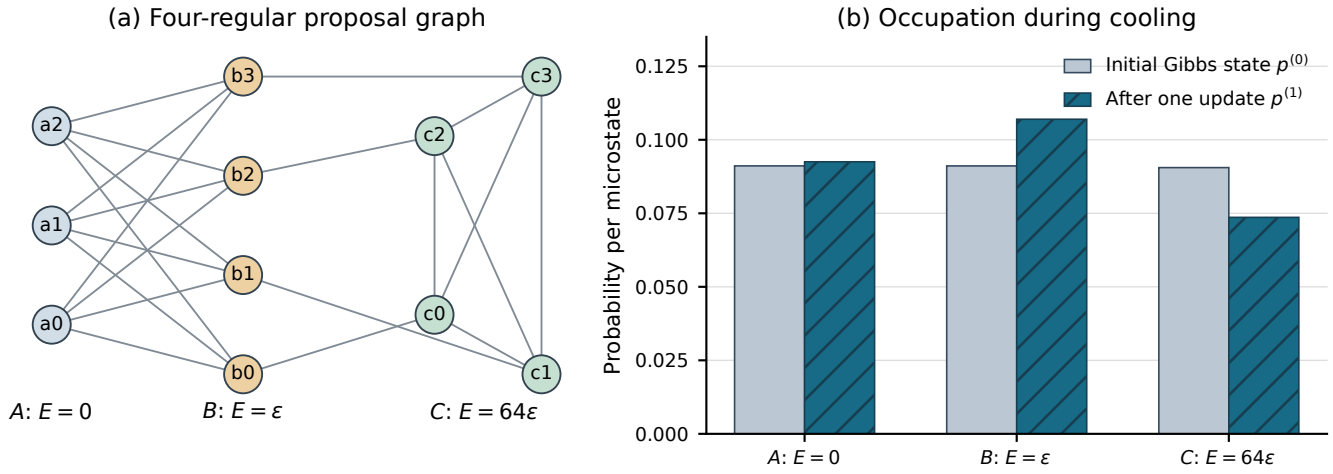


FIG. 1. (a) The complete proposal graph: 12 A–B edges, four matched B–C edges, and six edges inside C. Unmarked line crossings are not vertices, and the drawing is not an energy scale. (b) Per-microstate probabilities before and after the first update. The intermediate class B overtakes the lower-energy class A. Class sizes are 3, 4, and 4; heights are not class totals. The figure uses the specified graph and exact rational data, converted to floating point only for display, with drawing code drafted using OpenAI Codex (GPT-6).

Redefine the initial state to be Gibbsian for the perturbed energies at β_0. Tildes denote the resulting quantities. We prove the uniform bounds

$$\widetilde{\mathcal{A}}_1 > \frac{1003737}{2000000} > \frac{1}{2}, \qquad \widetilde{\overline{\mathcal{A}}}_2 > \frac{8014821}{16000000} > \frac{1}{2}. \tag{20}$$

For clarity write $P_k = P(\beta_k)$ and let a_k be its vector of escape probabilities. The function $f(q) = (1+e^q)^{-1}$ satisfies $|f'(q)| \le 1/4$. An edge energy difference in units of ϵ changes by at most 2δ. Summing the off-diagonal changes in a column, and then including the compensating diagonal change, yields

$$\|\widetilde{P}_k - P_k\|_1 \le b_k\delta, \qquad \|\widetilde{a}_k - a_k\|_\infty \le \frac{b_k\delta}{2}. \tag{21}$$

Here the matrix norm is the induced 1-norm. The ratio of perturbed to unperturbed initial Gibbs weights, after normalization, lies between $e^{-2b_0\delta}$ and $e^{2b_0\delta}$. Hence

$$\|\widetilde{p}^{(0)} - p^{(0)}\|_1 \le \eta_0, \qquad \eta_0 = e^{2b_0\delta} - 1. \tag{22}$$

By stochastic contraction, the 1-norm distance between the two distributions after the first update is at most $\eta_0 + b_1\delta$. Using $0 \le a_{k,i} \le 1$ in the activity inner products gives

$$\begin{aligned} |\widetilde{\mathcal{A}}_0 - \mathcal{A}_0| &\le \eta_0 + b_1\delta/2, \\ |\widetilde{\mathcal{A}}_1 - \mathcal{A}_1| &\le \eta_0 + b_1\delta + b_2\delta/2. \end{aligned} \tag{23}$$

These bounds hold on the full perturbation box, without retaining layer symmetry or truncating a perturbation series.

For the protocol (9),

$$b_0 < \frac{1}{10000}, \qquad b_1 < \frac{1}{32}, \qquad b_2 < \frac{21}{5}. \tag{24}$$

The first two bounds follow from $\log(1+q) < q$ for $q > 0$. For the third, $e^{7/10} > 1+7/10+(7/10)^2/2+(7/10)^3/6 > 2$. The inequality $e^q - 1 \le q/(1-q)$ for $0 \le q < 1$ also implies $\eta_0 < 1/4000000$. Substitution into Eq. (23) gives

$$\begin{aligned} |\widetilde{\mathcal{A}}_1 - \mathcal{A}_1| &< \frac{4263}{2000000}, \\ |\widetilde{\overline{\mathcal{A}}}_2 - \overline{\mathcal{A}}_2| &< \frac{17179}{16000000}. \end{aligned} \tag{25}$$

Subtracting these error bounds from 63/125 and 251/500, respectively, proves Eq. (20). Thus fully nondegenerate energies within the stated box also give counterexamples.

IV. MECHANISM AND SCOPE

The energy gaps ϵ and 63ϵ respond differently to the first temperature change. Probability flows from C into B while reverse $A \to B$ acceptance is still appreciable. The resulting $y > x$, visible in Fig. 1, reverses the per-state energy ordering on the A–B edges. The second cooling step makes $B \to A$ strongly favorable. Together with the remaining C occupation and its internal microstate jumps, this makes the full correction in Eq. (5) negative. The construction shows reachability from an exact Gibbs initial state; merely choosing an arbitrary inverted distribution would not answer the annealing question.

The equilibrium identity (6) and the explicitly hypothesized neighborhood-passivity sufficient condition of Ref. [1], Eq. (41), remain valid. Our result establishes that regularity, instantaneous detailed balance, a Gibbs initial state, and monotone cooling do not by themselves enforce the one-half bound. The same example rules out its two-update averaged counterpart. The latter mat-

ters for the activity substitution in the annealing performance analysis of Ref. [2], p. 4, in the discussion leading to Eq. (10).

This is a complete finite classical state space, with no state-space truncation, continuous-time normalization, extra laziness, or infinite-volume limit. The protocol has two finite temperature changes. The graph is not a single-spin-flip hypercube. No conclusion follows here for a restriction to the SK model, to hypercube proposal graphs, or to sufficiently slow cooling. Nor does failure of one intermediate activity bound alone disprove the final performance formulas, which involve additional estimates. We make no assertion of a minimal state count or maximal violation.

V. CALCULATION AND DATA AVAILABILITY

The complete 11×11 transition matrices were propagated in rational arithmetic and compared with a separately implemented sum of synchronous undirected-edge currents. A further independent implementation enumerated all $11 \times 8^2 = 704$ two-update proposal/acceptance histories, retaining distinct rejected proposals. All three calculations agree. These are exact finite calculations, not Monte Carlo estimates.

As a convention check, the equilibrium four-spin periodic ferromagnetic Ising chain with $H = -J \sum_{i=1}^{4} s_i s_{i+1}$, $s_5 = s_1$, and $\beta J = (\log 2)/2$ gives $Z = 41/2$ and acceptance $16/41$ by enumeration. This agrees with the finite-chain formula of Burovski *et al.* [3], arXiv v2, Appendix, subsection B, Eq. (24), evaluated at $L = 4$ and $e^{2\beta J} = 2$. The value is a substitution into that published formula, not an additional result of the present work.

OpenAI Codex was used substantively to propose and analyze the construction, write exact-arithmetic code, and assist source comparisons. The automated work was directed by fixed requirements for the Glauber rule, Gibbs initialization, admissible cooling, and exact sign checks. Verification consisted of the distinct calculations above and analytic error bounds; AI agreement alone was not used as a sign certificate. The manuscript and plotting-code preparation used GPT-6. Exact model-version identifiers for earlier research stages have not been recovered in this draft. Final human scientific review and confirmation of the disclosure remain pending.

All defining inputs and the scalar sign-verification algorithm are in this article. The accompanying local source package contains the full propagation, edge-current and history-enumeration implementations, rational outputs, and regeneration instructions. No experimental data are used. A public repository has not been deposited for this draft.

ACKNOWLEDGMENTS

OpenAI Codex (GPT-6 for manuscript preparation) assisted with drafting and editing the text, bibliography preparation, and presentation code. This disclosure includes substantive research assistance described above and does not represent completed human review.

Appendix A: A short exact sign certificate

The following Python 3 code uses only the standard library. It evaluates the displayed scalar formulas and checks every enclosure in Table I. Integer cross multiplication implements comparisons of `Fraction` objects, so no numerical precision setting is needed.

```
from fractions import Fraction as F
r = F(10000, 10001); s = r**64
Z = 3 + 4*r + 4*s
u, v, w = 1/Z, r/Z, s/Z
h = F(33, 65)
ell = F(33**63, 33**63 + 32**63)
j = (h*v - (1-h)*u)/4
L = (ell*w - (1-ell)*v)/4
x, y, t = u + 4*j, v - 3*j + L, w - L
a0 = 3*u*(1-h) + v*(3*h + 1-ell)
a0 += w*(F(3, 2) + ell)
rho = F(1, 1 + 64**63)
a1 = 3*x/65 + (F(192, 65) + rho)*y
a1 += (F(5, 2) - rho)*t
vals = [a0, a1, (a0+a1)/2, x, y, t]
nums = [499785676488, 504349849005,
        502067762747, 92521064942,
        106989212896, 73619988396]
for a, m in zip(vals, nums):
    assert F(m, 10**12) <= a
    assert a < F(m+1, 10**12)
assert a1 > F(63, 125)
assert (a0+a1)/2 > F(251, 500)
```

[1] Y. Luo, Y.-Z. Zhen, X. Liu, D. Ebler, and O. Dahlsten, Bound on annealing performance from stochastic thermodynamics, with application to simulated annealing, Phys. Rev. E **108**, 054119 (2023), equation references in the text refer to the v1 author preprint, arXiv:2311.10424v1 [cond-mat.stat-mech].

[2] Y. Luo, Y.-Z. Zhen, X. Liu, D. Ebler, and O. Dahlsten, General limit to thermodynamic annealing performance, Phys. Rev. E **108**, L052105 (2023), page and equation references refer to the v1 author preprint, arXiv:2311.10422v1 [cond-mat.stat-mech].

[3] E. Burovski, W. Janke, M. Guskova, and L. Shchur, Acceptance rate is a thermodynamic function in local Monte Carlo algorithms, Phys. Rev. E **100**, 063303 (2019), arXiv:1907.06776v2 [cond-mat.stat-mech].